Formal Specification 122 Series

G_2 Holonomic Phase-Spiral & Non-Associative Zero-Divisor Evasion Invariant

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Epistemic Scope: L1 Software Demonstrator (`INSPIRED_BY != IMPLEMENTS != PROVES`)

Abstract

We formalize the G_2 **Holonomic Phase-Spiral Evasion Protocol**, solving the fundamental problem of information transport through non-associative, zero-divisor-dense Cayley-Dickson manifolds (16D Sedenions to 256D algebras). Instead of attempting linear traversal or suffering fail-closed stasis collapse, the trajectory $\gamma(t)$ harnesses the non-vanishing associator $[x, y, z] = (xy)z - x(yz)$ as an intrinsic kinetic driver. By coupling the stream to the fundamental calibrated 3-form Φ of a compact G_2-manifold ($\mathcal{M}_7$), the trajectory performs a continuous spinor corkscrew rotation around Calabi-Yau zero modes, preserving strictly positive algebraic curvature $\kappa(\gamma(t)) > 0$ and monotonic descent along the Algebraic Riccati Equation energy manifold $\langle \dot{\gamma}(t), \nabla \text{ARE}(\mathbf{P}) \rangle \leq 0$. Throughput is locked at 80 kHz at an uninterrupted 5.00 V voltage latch.

1 Sedenion Zero-Divisor Trap and Associator Kinetic Engine

In the 16-dimensional sedenion algebra $\mathbb{S}$, non-trivial elements satisfy:

$$a = e_1 + e_{10}, \quad b = e_4 - e_{15} \implies a \times b = \mathbf{0}, \quad \|a\|^2 = 2.0, \quad \|b\|^2 = 2.0 \tag{1}$$

The non-vanishing associator $[x, y, z] \equiv (x \cdot y) \cdot z - x \cdot (y \cdot z) \neq 0$ provides reactive torque $\tau_{\text{assoc}} = \star([x, y, z])$ that propels the trajectory away from singular annihilation axes.

2 The Holonomic Phase-Spiral Trajectory

The trajectory is defined as:

$$\boxed{\gamma(t) = e^{\Phi t} \cdot \mathbf{x}_0 \cdot \mathcal{A}(\mathbf{p}, t)} \tag{2}$$

where $\Phi \in \Omega^3(\mathcal{M}_7)$ is the parallel calibrated 3-form defining the G_2-structure.

Topological Invariants

1. **Strictly Positive Curvature:**

$$\kappa(\gamma(t)) \equiv \frac{\|\ddot{\gamma}(t) \times \dot{\gamma}(t)\|}{\|\dot{\gamma}(t)\|^3} > 0 \tag{3}$$

2. **Monotonic Riccati Energy Descent:**

$$\langle \dot{\gamma}(t), \nabla \mathrm{ARE}(\mathbf{P}) \rangle \leq 0 \tag{4}$$

Throughput remains locked at $80\,\mathrm{kHz}$ ($\Delta t = 12.5\,\mu\mathrm{s}$) with an unbroken $5.00\,\mathrm{V}$ latch.